

ME 300: Engineering Thermodynamics

Pressure, Temperature, and Density



Mass, moles, and volume

Mass: how much material (M) - kg

Moles: quantity of molecules (N) $\rightarrow$ $1 \text{ mole} = 6.02214199 \times 10^{23} \text{ part.}$

Volume: amount of space occupied (V) - m^3

Density and specific volume

$$\rho = \frac{M}{V} \sim \left[\frac{\text{kg}}{\text{m}^3} \right]$$

define ρ at a point



$$\Delta x, \Delta y, \Delta z \rightarrow 0$$

→ continuum: $\Delta V = \Delta x \Delta y \Delta z$

large number of particles inside

→ Knudsen number = λ / L → λ = mean free path
 L = dimension of system

$$L \gg \lambda$$

→ mass-specific volume: $v = \frac{V}{M} = \frac{1}{\rho} \sim \left[\frac{\text{m}^3}{\text{kg}} \right]$

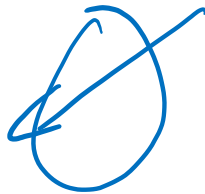
molar-specific volume $\Rightarrow \bar{V} = \frac{V}{N} \sim \left[\frac{\text{m}^3}{\text{mol}} \right]$

Pressure

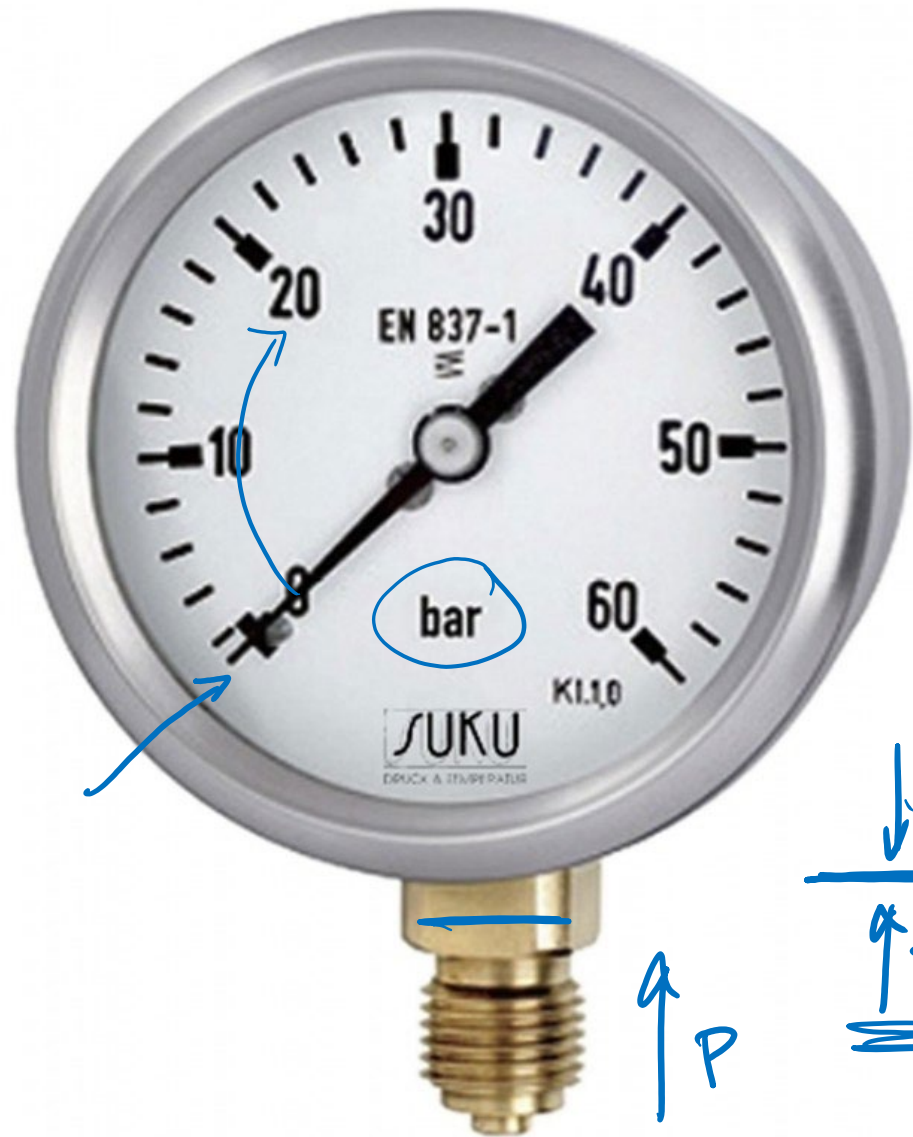
→ normal force exerted by fluid on an area

$$P = \frac{F}{A}$$

→ intensive: $P = \lim_{\Delta A \rightarrow 0} \frac{F_{\text{normal}}}{\Delta A}$

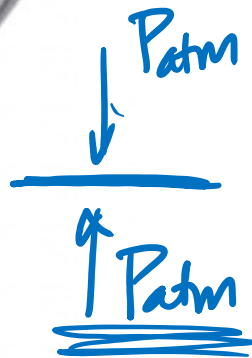
→  flux of molecules ΔA

Gage Pressure



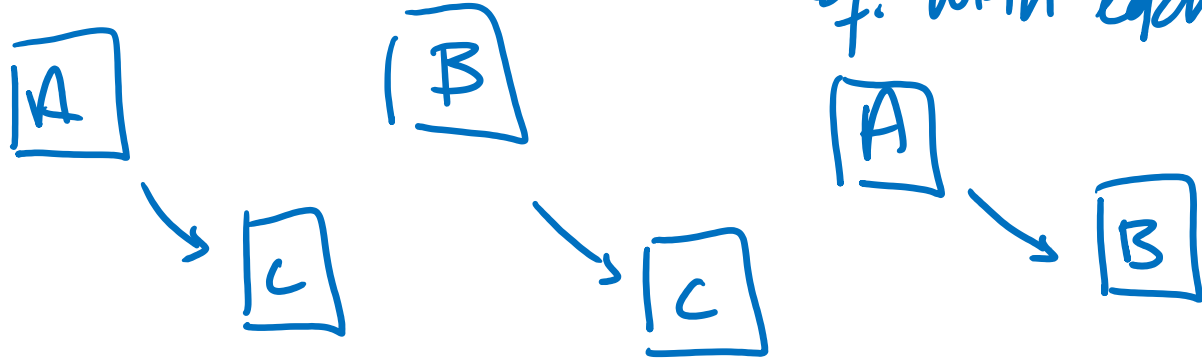
$$P_{abs} = P_{gage} + P_{atm}$$

↑ ↑



Temperature

Zeroth law of thermodynamics: if two systems are in thermal equilibrium with a third sys., then they are in thermal eq. with each other



$\Rightarrow T$ measure of the mean molecular speed